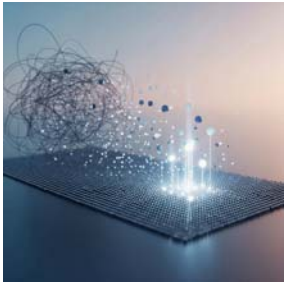


A4.2.2 Feature selection



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After data has been cleaned, the next question is which features, which columns in the dataset, should actually be given to the model. A dataset may contain dozens or hundreds of columns, but many will be irrelevant, redundant, or actively harmful to the model's ability to generalise. Feature selection is the process of identifying and retaining only the most informative attributes. This page covers why feature selection matters, the three strategies used to perform it, and how to choose between them.

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Key Questions

- What is feature selection and how does it differ from the irrelevant data removal covered in A 4.2.1?
- What are the trade-offs between filter, wrapper, and embedded methods in terms of computational cost, accuracy of selection, and model dependency?
- What are the three feature selection strategies, what named techniques does each include, and when does each perform its selection relative to model training?
- Given a scenario, how would you decide which strategy is most appropriate?

IB Command Terms

Describe

Give a detailed account of the features or characteristics. For feature selection, this means explaining the role of the process, describe each strategy with its mechanism, name specific techniques within each strategy, and explain the trade-offs.

What to include

For each strategy make sure to include when the selection occurs, how features are evaluated, the names of at least one specific technique, the computational cost, and when it is preferred.

An answer that only names the three strategies without describing how each works will not score full marks.

The role of feature selection

In machine learning, the data provided to a model is represented as a set of features, the columns of a dataset, each describing one attribute of an example. A model that predicts house prices might have features such as square footage, number of bedrooms, age, postcode, and if the data was not carefully cleaned the previous owner's name and a unique listing reference.

Not all features contribute useful information. Some are irrelevant because they have no logical or statistical relationship with the target variable. Others are redundant because they contain the same information as another feature already in the dataset.

Including these features does not help the model; it harms it.

REDUCES OVERFITTING RISK	REDUCES TRAINING TIME	IMPROVES INTERPRETABILITY
A model with too many features can memorise coincidental patterns in the training data that do not generalise to new data. This is called overfitting. When a model overfits, it performs well on training data but poorly on any new example. Removing irrelevant features reduces the number of spurious patterns the model can latch onto.	Each feature the model processes adds computation. With 10 features, training is fast. With 10,000 features, training may take hours or days longer than necessary. Removing features that contribute no useful information reduces computation without reducing model quality.	A model built on 5 well-chosen features is easier to explain and audit than one built on 500. In regulated domains such as medicine, finance, and law, being able to explain why a model made a prediction is often a requirement. Feature selection makes that possible.

CONNECTION TO A 4.2.1

Feature selection builds on the irrelevant data removal step covered in A 4.2.1.

In A 4.2.1, obviously irrelevant columns (such as a unique ID or a phone number) are removed by inspection. Feature selection goes further as it uses statistical tests or model-based evaluation to identify features that are subtly irrelevant or redundant.

Redundant features are ones that might look plausible but do not actually help the model make better predictions.

Feature Selection Strategies

There are three main strategies for selecting features. The fundamental distinction between them is when the selection happens relative to model training, and whether the selection process involves the model itself.

Filter methods - selection occurs before model training

When: Before any model is trained. Filter methods treat feature selection as an independent preprocessing step, separate from the learning algorithm.

How it works: Each feature is scored individually using a statistical measure of its relationship with the target variable. Features above a defined threshold score are retained; those below are removed. Because features are scored independently, the method is fast and does not require a model to be trained.

Statistical tests used: Correlation coefficient (Pearson): measures the linear relationship between a numerical feature and a numerical target. Higher absolute value means stronger relationship. Chi-squared test: used when both the feature and target are categorical. Measures how much the observed distribution of the feature differs from what would be expected if the feature were independent of the target. Mutual information: measures how much knowing the value of a feature reduces uncertainty about the target. Works for both numerical and categorical data.

Named techniques: Correlation-based feature selection; chi-squared feature selection; mutual information scoring; variance threshold (removes features with near-zero variance, a feature that is the same value for almost every example provides no information).

Advantages: Very fast as no model training required. Computationally inexpensive and easy to interpret. Works well as an initial pass to remove obviously uninformative features.

Key limitation: Features are scored independently. A feature that is individually weak may be highly informative when combined with another feature, filter methods cannot detect this interaction. They may also retain redundant features that are both highly correlated with the target but also highly correlated with each other.

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ANALOGY

A librarian asked to select the 20 most useful books from a collection of 500 reads the title and summary of each book individually and scores it. They do not consider how the books relate to each other, they select the individually highest-scoring titles. This is fast but may result in selecting five books on the same narrow topic when one would suffice.

Wrapper methods - selection occurs around model training

When: Around training. The model is used as part of the selection process, features are evaluated not by a statistical test but by how well a model trained on them actually performs.

How it works: A subset of features is selected, a model is trained and evaluated on that subset, and the performance is recorded. The process is repeated for many different subsets. The subset that

produced the best-performing model is selected. Because the model must be trained for each candidate subset, this is computationally expensive, the number of possible subsets grows exponentially with the number of features.

Named technique 1: Forward selection: Begin with no features. Add features one at a time, each time choosing the feature that most improves model performance. Continue until adding another feature no longer improves the model above a threshold.

Named technique 2: Backward elimination: Begin with all features. Remove features one at a time, each time removing the feature whose removal least degrades model performance. Continue until further removal causes performance to drop below a threshold.

Named technique 3: Recursive feature elimination (RFE): Train the model on all features. Rank features by their importance to the model. Remove the least important feature. Retrain. Repeat until the desired number of features remains. RFE is the most common wrapper method in practice and is available in standard ML libraries.

Advantages: Finds the combination of features that genuinely works best for the specific model being used. Captures feature interactions that filter methods miss.

Key limitation: Computationally expensive as training the model repeatedly for each subset is slow for large feature sets. The selected features are specific to the chosen model and may not transfer well if the model is changed.

ANALOGY

An optometrist determines the correct prescription by testing many combinations of lenses, asking the patient which combination produces the clearest vision. Each test involves actually looking through the lenses (training the model). The result is highly accurate for that specific patient (model) but the process takes much longer than simply measuring the eye independently of any glasses.

Embedded methods - selection occurs during model training

When: During training. Some learning algorithms are designed to evaluate feature importance as part of their training process. Feature selection is a by-product of learning, not a separate step.

How it works: As the model adjusts its parameters during training, it also assigns an importance weight to each feature. Unimportant features receive weights close to zero. After training, features with very low weights can be identified and removed. Because this happens during the single training run, there is no additional computational cost.

Named technique 1: Lasso regression (L1 regularisation): Lasso adds a penalty to the training objective that is proportional to the absolute value of each model coefficient. This penalty encourages the algorithm to shrink the coefficients of unimportant features to exactly zero, effectively removing them from the model. Features with a non-zero coefficient after training are the selected features. Lasso is the canonical embedded feature selection method.

Named technique 2: Decision tree feature importance: Decision trees and ensemble methods such as Random Forests split the data at each node using the feature that most reduces impurity (uncertainty about the target). The frequency with which a feature is used for splits, and the improvement it provides, gives a natural measure of its importance. Low-importance features are rarely or never used for splits.

Advantages: No additional computational cost beyond model training. The selection is directly tailored to the model and the learning objective. Works well for models where embedded selection is a natural part of the algorithm.

Key limitation: Only available for algorithms that support it because not all models produce feature importance scores. The importance measure is specific to the algorithm and may not generalise to other model types.

ANALOGY

A chef follows a recipe that automatically records which ingredients were used most often and in what quantities across many meals. After cooking, the recipe highlights the ingredients that consistently appeared in the most successful dishes. The selection happened during cooking, not before.

COMMON EXAM MISTAKE

Lasso regression is a named embedded method and should be mentioned explicitly in any exam answer about embedded feature selection. Stating only that 'the model assigns importance weights to features' without naming a specific algorithm will not achieve full marks. Similarly, for wrapper methods, naming forward selection, backward elimination, or RFE is expected. 'The model is trained on subsets' as an answer alone is insufficient.

Comparing the three strategies

The table below summarises the key characteristics of each strategy. In an exam question asking you to recommend a strategy for a given scenario, you should use this framework to justify your choice.

STRATEGY	WHEN SELECTION OCCURS	NAMED TECHNIQUES	COMPUTATIONAL COST	SPEED	KEY LIMITATION
Filter methods	Before training - independent of any model	Statistical tests (correlation, chi-squared, mutual information)	Low - scores each feature individually	High - no model training required	<i>May select features that are individually correlated but redundant together</i>
Wrapper methods	Around training - uses a model to evaluate subsets	Forward selection; backward elimination; recursive feature elimination (RFE)	Very high - trains a model for each candidate subset	Low - can be prohibitive with many features	<i>Finds the best combination for the specific model; can overfit the selection to the training set</i>
Embedded methods	During training - built into the algorithm	Lasso regression (L1 regularisation); decision tree feature importance	Low - selection happens as a by-product of training	High - no extra training runs needed	<i>Only available for algorithms that support it; less interpretable than filter methods</i>

HOW TO CHOOSE A STRATEGY

Use **filter methods** when the dataset has a very large number of features and you need a fast initial reduction; when computational resources are limited; or as a first pass before applying a more expensive method.

Use **wrapper methods** when the model performance must be maximised and computational cost is acceptable; when feature interactions are likely to matter; or when you are using a model for which no embedded method exists.

Use **embedded methods** when you are using an algorithm that supports them (such as Lasso or a decision tree); when you want selection to happen automatically without a separate step; or when computational efficiency is a priority alongside accuracy.

Worked example: applying filter methods to a house price dataset

A data scientist is building a supervised regression model to predict property sale prices. The dataset has six features. A filter method using Pearson correlation is applied. Each feature is scored on its correlation with the target variable (sale price), where 1.0 means perfect positive correlation and 0.0 means no linear relationship.

A threshold of 0.5 is set: features scoring below 0.5 are removed.

FEATURE	DESCRIPTION	CORRELATION SCORE	DECISION	
square_footage	Total interior floor area in square metres	0.87	Keep	
num_bedrooms	Number of bedrooms	0.74	Keep	
age_of_property	Age of property in years	0.61	Keep	
postcode	Full postcode string	0.09	Remove	
listing_ref	Unique estate agent reference number	0.02	Remove	
prev_owner_name	Name of previous owner	0.00	Remove	

READING THE TABLE

square_footage (0.87): strong positive correlation with sale price - larger properties sell for more. Retained.

num_bedrooms (0.74): meaningful correlation - more bedrooms associated with higher prices. Retained.

age_of_property (0.61): moderate correlation - older properties tend to sell for less. Retained.

postcode (0.09): very low correlation - while location matters, the raw postcode string is not a numerical feature and does not correlate linearly with price in this form. Removed. (Note: postcode would typically be encoded differently before modelling.)

listing_ref (0.02): near-zero correlation - a unique identifier has no relationship with price. Removed.

prev_owner_name (0.00): zero correlation - entirely irrelevant. Removed.

After filter selection, the model is trained on three features rather than six. Training is faster, and the risk of overfitting to listing references or owner names is eliminated.

THINK FIRST

After running filter selection, a colleague suggests also applying a wrapper method (recursive feature elimination) to the three retained features. What would be the benefit of doing this, and what would be the cost?

ANSWER TO THINK FIRST

Benefit: RFE would evaluate which of the three retained features genuinely improves model performance when used in combination. It is possible that age_of_property adds little beyond what square_footage already captures, RFE would detect this redundancy and might recommend training on only two features.

Cost: RFE requires training the model multiple times (once for each candidate subset). For only three features this is trivial, but the principle matters, wrapper methods become expensive when the feature count is large.

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

What is the primary goal of feature selection?

- A. ☐ To ensure the data is clean.

B. ☐ To make the dataset larger.

C. ☒ To select the most informative attributes to improve model performance.

D. ☐ To add more complexity to the model.

2

A model that learns from irrelevant noise in the data instead of the true underlying patterns is said to be:

A. ☒ Overfitting

B. ☐ A wrapper method

C. ☐ Underfitting

D. ☐ Standardized

3

Which feature selection strategy evaluates subsets of features using a machine learning model in a trial-and-error process?

A. ☐ Embedded Methods

B. ☐ Imputation

C. ☒ Wrapper Methods

D. ☐ Filter Methods

4

A data scientist uses a statistical correlation test to score and select features before training any model. This is an example of:

A. ☐ An Embedded Method

B. ☐ A Wrapper Method

C. ☒ A Filter Method

D. ☐ Normalization

5

Which of the following is NOT a benefit of performing feature selection?

A. ☐ Reduced training time

B. ☐ Reduced risk of overfitting

C. ☒ Increased number of features

D. ☐ Improved model accuracy

6

The analogy of "sifting flour before baking a cake" best describes which feature selection strategy?

A. ☐ Wrapper Methods

B. ☒ Filter Methods

C. ☐ Data Cleaning

D. ☐ Embedded Methods

7

Which feature selection strategy is generally the most computationally expensive and time-consuming?

A. ☐ Embedded Methods

B. ☒ Wrapper Methods

C. ☐ Filter Methods

D. ☐ Standardization

8

Some algorithms, like Lasso Regression, naturally perform feature selection during the training process. This is an example of:

A. ☐ A Filter Method

B. ☐ An Outlier

C. ☒ An Embedded Method

D. ☐ A Wrapper Method

9

A data scientist is building a model to predict house prices. Which of the following features is the best candidate to be removed during feature selection?

A. ☐ Number of bedrooms

B. ☐ Square footage

C. ☐ Age of the house

D. ☒ The previous owner's name

10

The key difference between the three main feature selection strategies is:

A. ☐ The type of data they work on.

B. ☐ When the feature selection occurs relative to the model training process.

C. ☐ The programming language used.

D. ☐ The number of features they can handle.

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